PTC
多元微积分
期末资料

PTC Finals Prep Materials for Multivariable Calculus

PTC 伴学社

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MTH004&MTH008

前言　preface

致每一位拿到这份资料的同学：

欢迎阅读 PTC 多元微积分期末复习资料！

本册资料为专题知识点复习，我们的资料根据所学章节按专题展开，分为专题词汇、基本思想与实战演练三部分，在此基础上加入了我们自己的辅助理解与解题思路，能够帮助同学在巩固基础的同时加强对知识的理解、习题的掌握。

这份期末资料经过了反复打磨，最后精简成了你手中的这份资料。这离不开我们资料组每一位成员的付出，使我们的资料得以成形；也得益于 PTC 每一位成员的帮助，使我们资料的精准度和完美度不断提高。

预祝各位阅读愉快，复习顺利，在期末考试中取得惊喜的成绩！

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（按拼音首字母排序，顺序不分先后）

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点乘与叉乘

专题数学词汇：Origin 原点
Octant 卦限
Cross product (= vector product) 叉积，向量积
Anti-commutative law 反交换律
Equality 等式
Adjoin two equations 把两个等式联接起来
Identify 恒等式
Lagrange's Identity 拉格朗日恒等式
Principal square root 主平方根, 非负平方根
Parallelepiped 平行六面体
Edge 边
Intrinsic property 内在的、固有的性质
Trivially 平凡地
Adjacent sides 相邻边
Cyclic order 循环次序
Coincide with 重合

【基本思想】

1. 点乘 Dot product (Scalar product)

①定义

$$\mathbf{u}\cdot\mathbf{v}=\langle u_1,u_2,u_3\rangle\cdot\langle v_1,v_2,v_3\rangle=u_1v_1+u_2v_2+u_3v_3$$

②性质

If **u**, **v** and **w** are vectors, and c is a scalar, then

$\mathbf{u}\cdot\mathbf{v}=\mathbf{v}\cdot\mathbf{u}$ $\qquad$ $\mathbf{u}\cdot(\mathbf{v}+\mathbf{w})=\mathbf{u}\cdot\mathbf{v}+\mathbf{u}\cdot\mathbf{w}$

$c(\mathbf{u}\cdot\mathbf{v})=(c\mathbf{u})\cdot\mathbf{v}$ $\qquad$ $0\cdot\mathbf{u}=0$ $\qquad$ $\mathbf{u}\cdot\mathbf{u}=\|\mathbf{u}\|^2$

③几何意义

If θ is the smallest nonnegative angle between the nonzero vectors **u** and **v**,

then $u\cdot v=\|u\|\|\mathbf{v}\|\cos\theta$

④点乘的常见应用

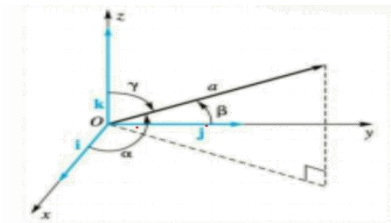
a.Computing lengths and Angles

$$\cos\theta=\frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|},\ \theta\in[0,\pi]$$

b.Determine whether two vectors are perpendicular

$\mathbf{u}\perp\mathbf{v}\leftrightarrow\mathbf{u}\cdot\mathbf{v}=0$

c.Direction Angles and Cosines



If $a=a_1i+a_2j+a_3k$, then

$\cos\alpha=\frac{\boldsymbol{\alpha}\cdot\mathbf{i}}{\|\boldsymbol{\alpha}\|\|\mathbf{i}\|}=\frac{\mathbf{a_1}}{\|\alpha\|}$ $\cos\beta=\frac{\boldsymbol{\alpha}\cdot\mathbf{j}}{\|\alpha\|\|\mathbf{j}\|}=\frac{a_2}{\|\boldsymbol{\alpha}\|}$ $\cos\gamma=\frac{\boldsymbol{\alpha}\cdot\mathbf{k}}{\|\boldsymbol{\alpha}\|\|\mathbf{k}\|}=\frac{a_3}{\|\alpha\|}$

$\cos2\alpha+\cos2\beta+\cos2\gamma=1$

d.Projection of **u** on **v**

Vector projection:

pr**vu** = $(\frac{u\cdot v}{\|\mathbf{v}\|^2})\mathbf{v}=\|\mathbf{u}\|\cos\theta\ \frac{v}{\|v\|}$ (求做功 W 也会用到这个式子)

Scalar projection: $\|u\|\cos\theta=\frac{u\cdot v}{\|v\|}$

【Tips】 题目中只说 projection 则为求 vector projection, 求 scalar projection 会特别指出

2.叉乘 Cross product (Vector Product)

①定义

$$u\times v=\begin{vmatrix}u_2 & u_3\\ v_2 & v_3\end{vmatrix}i-\begin{vmatrix}u_1 & u_3\\ v_1 & v_3\end{vmatrix}j+\begin{vmatrix}u_1 & u_2\\ v_1 & v_2\end{vmatrix}k=\begin{bmatrix}i & j & k\\ u_1 & u_1 & u_1\\ v_1 & v_1 & v_1\end{bmatrix}$$

②性质

If **u**, **v**, and **w** are vectors in three-space and k is a scalar,then:

1. **u** × **v** = -(**v** × **u**) (anticommutative law)
2. **u** × (**v** + **w**) = (**u** × **v**) + (**u** × **w**)
3. k (**u** × **v**) = (k**u**) × **v** = **u** ×(k**v**)
4. **u** × **0** = **0** × **u** = **0** ; **u** × **u** = **0**
5. (**u** × **v**)·**w** = **u**·(**v** × **w**)
6. **u** × (**v** × **w**) = (**u**· **w**)**v** - (**u**· **v**) **w**=（**u**×**v**) × **w**+**v** × (**u**×**w**)

③几何意义

u·（u × v）= 0 = v·（u × v), that is, u × v⊥u and u × v⊥v

‖**u** × v‖ = ‖u‖ ‖v‖ sinθ

④叉乘的常见应用

a.Determine whether two vectors u and v are parallel

u//v ↔ u ×v= 0（与点乘区别，一个证垂直一个证平行）

b.Find a vector perpendicular to both u and v

c.Find the area of a parallelogram with u and v as adjacent sides.

Area= ‖u× v‖

d.Find the volume of the parallelepiped determined by the vectors with a, b, c

$$V=|a\cdot(b\times c)|=\left\|\begin{matrix}a_1 & a_2 & a_3\\ b_1 & b_2 & b_3\\ c_1 & c_2 & c_3\end{matrix}\right\|$$

【实战演练】

【Example 1】

a=(1,2,3),b=(2,4,6),c=(5,0,2)

Please calculate follow

(1)a×b (2) ‖ a×c ‖ (3)if (a×c)⊥b? (4)(a·b)×c

【Solution】

(1)a×b=$\begin{vmatrix} i & j & k \\ 1 & 2 & 3 \\ 2 & 4 & 6 \end{vmatrix}$=0

注意:a 与 b 平行但叉乘为零，不能与垂直点乘为零混淆

(2)a×c=$\begin{vmatrix} i & j & k \\ 1 & 2 & 3 \\ 5 & 0 & 2 \end{vmatrix}$=$\begin{vmatrix} 2 & 3 \\ 0 & 2 \end{vmatrix} i - \begin{vmatrix} 1 & 3 \\ 5 & 2 \end{vmatrix} j + \begin{vmatrix} 1 & 2 \\ 5 & 0 \end{vmatrix} k$= 〈4,13,-10〉

‖ a×c ‖=$\sqrt{4^2 + 13^2 + 10^2}$=$\sqrt{285}$

(3)(a×c)⊥b so we can calculate (a×c)·b= 〈4,13,-10〉〈2,4,6〉=0, they are vertical.

(4)(a·b)×c= 〈2,8,18〉 × 〈5,0,2〉 =$\begin{vmatrix} i & j & k \\ 2 & 8 & 18 \\ 5 & 0 & 2 \end{vmatrix}$

=$\begin{vmatrix} 8 & 18 \\ 0 & 2 \end{vmatrix} i - \begin{vmatrix} 2 & 18 \\ 5 & 2 \end{vmatrix} j + \begin{vmatrix} 2 & 8 \\ 5 & 0 \end{vmatrix} k$= 〈16,86,-40〉

空间平面方程与直线方程综合应用

专题数学词汇：Accumulated arc length 累积弧长
Calculus of vector functions 向量微积分/向量分析
Clockwise 顺时针方向
Counterclockwise 逆时针
Concave side 曲线凹向一侧
Secant line 割线
Tetrahedron 四面体
Symmetric equations 对称式方程

一．Find the equation of the line 求解直线方程

【基本思想】

Lines in Three-Space 三维空间的直线

1.Parametric equations 参数方程

定义：The **parametric equations** of the line through the point P$(x_0, y_0, z_0)$and parallel to **v**=<a,b,c>(direction vector) are,

$$\begin{cases} x = x_0 + at \\ y = y_0 + bt \\ z = z_0 + ct \end{cases}, t \in R$$

The numbers a, b, and c are called **direction numbers** for the line.

Tips:Vector equation and Parametric equation for a line are NOT unique.

2. Symmetric equations 对称式方程

定义：The **symmetric equations** of the line through the point P(x_0, y_0, z_0) and parallel to **v**=<a,b,c>(direction vector) are,

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}$$

3. Two-points Form 两点式方程

Two points determine a straight line.

$$\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1} = \frac{z - z_1}{z_2 - z_1}$$

4. Standard Form 一般方程

Two intersecting planes determine a straight line.

$$\begin{cases} A_1x + B_1y + C_1z + D_1 = 0 \\ A_2x + B_2y + C_2z + D_2 = 0 \end{cases}$$

【实战演练】

【Example 1】

Find the symmetric equations of the line through (4, 0, 6) and perpendicular to the plane x - 5y + 2z = 10.

【Solution】$\frac{x-4}{1} = \frac{y}{-5} = \frac{z-6}{2}$

二．Find the Tangent line to a curve 求曲线的切线方程

【解题模板】
先对向量值函数在某一点上求导，即为切线的方向向量，再依据题干要求，依据其他信息，写出对应切线（直线）方程.
【实战演练】
【Example 1】
Find the parametric equations of the tangent line to the curve x = $2t^2$, y = 4t, z =t^3 at t = 1.

【Solution】$\begin{cases} x = 2 + 4t \\ y = 4 + 4t \\ z = 1 + 3t \end{cases}$

三,Find the equation of the plane 求平面方程
【基本思想】
Planes in Three-Space 三维空间的平面
1. 标准方程

The Standard Equation of a plane is $Ax + By + Cz = 0$

2. 点法式方程
The equation of the plane through the point P(x_0, y_0, z_0) and perpendicular to $\mathbf{n} =< A, B, C >$ is

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$$

3. 截距式方程
The equation of the plane through the point P(a,0,0),Q(0,b,0),R(0,0,c)is,

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

【解题模板】
平面上不共线三点，确定一个平面：
解法 1：构造平面上两个向量，进行叉积运算得到法向量，得到点法式.
解法 2：设第四点 M(x, y, z) 与其余三点共面，构造平面上三个向量，进行混合积运算，利用行列式为零构造方程.
【实战演练】
【Example 1】

Let Ⅱ be the plane passing through (3,1,−2) with normal vector <11,9,-5>.

Find the equation of Ⅱ.

【Solution】
The equation of the plane is 11x+9y−5z=52.
【Example 2】
Find the equation of the plane containing the line x = 1 + 2t, y = -1 + 3t, z = 4 + t and the point (11, -1, 52).(Hit:Let t equals to particular values)
【Solution】

72x−43y−15z=55

三维空间中常见的距离问题

专题数学词汇：normal vector 法向量

direction vector 方向向量

projection 投影

parallel 平行

【基本思想】

1. 点到线

let P(x_0, y_0, z_0), straight line L through the point Q(x_1, y_1, z_1)，the direction vector is $\vec{v} = (a, b, c)$，there are two formulas for the distance from the point P to the line L：

a. Vector cross multiplication formula：$d = \frac{|\overrightarrow{QP} \times \vec{v}|}{|\vec{v}|}$, $\overrightarrow{QP} = (x_0 - x_1, y_0 - y_1, z_0 - z)$

b. Projection method：$d = \sqrt{|QP|^2 - \left(\frac{|\overrightarrow{QP} \cdot \vec{v}|}{|\vec{v}|}\right)^2}$

2. 点到面

let P(x_0, y_0, z_0)，The equation for the plane is β Ax + By + Cy + D = 0, its normal vector is $\vec{n} = (A, B, C)$，so

$$d = \frac{|Ax_0 + By_0 + Cz_0 + D|}{\sqrt{A^2 + B^2 + C^2}}$$

3. 线到面（线必须平行平面）

Let the direction vector of the line L be $\vec{v} = (a, b, c)$，a point M on a straight line (x_1, y_1, z_1)，the plane equation is β

Ax + By + Cy + D = 0 ，So the plane normal vector is $\vec{n} = (A, B, C)$，because the line is parallel to the plane, so

$\vec{n} \cdot \vec{v} = 0$ ，则 $d = \frac{|Ax_1 + By_1 + Cz_1 + D|}{\sqrt{A^2 + B^2 + C^2}}$

4. 面到面（面与面需平行）

Set the plane α Ax + By + Cy + D_1 = 0

the plane β Ax + By + Cy + D_2 = 0

$$d = \frac{|D_1 - D_2|}{\sqrt{A^2 + B^2 + C^2}}$$

【解题思路】

1. 弄清题目求谁和谁的距离
2. 如果是线与线，线与面或面与面，则需事先判断是否平行

【实战演练】

【Example 1】

Find the distance between the two lines x = 2 – t , y = 3 + 3t ,z = 2t and x = -1 + t,y = 2 – 3t , z = -1 – 2t

【Solution】

When t=0 , we get the point A(2,3,0) is on the first line and the plane P：-(x-2)+3(y-3)+2z = 0 is perpendicular to the two lines . The second line intersects the plane P at B $\left(\frac{-8}{7},\frac{17}{7},\frac{-5}{7}\right)$. The line segment AB is perpendicular to the two lines .So, the distance between A and B is the distance between the two parallel lines. The distance is

$$\sqrt{\left(2+\frac{8}{7}\right)^2+\left(3-\frac{17}{7}\right)^2+\left(\frac{5}{7}\right)^2}=\frac{5}{7}\sqrt{21}$$

求偏导数及高阶偏导数

专题数学词汇：partial derivatives 偏导数
higher partial derivatives 高阶偏导数

【基本思想】

1. $f_x(x,y) = \lim_{\Delta x \to 0} \frac{f(x+\Delta x,y)-f(x,y)}{\Delta x}$
2. 偏导数：多元函数中其他变量保持不变的情况下，函数对某一自变量的变化率
3. 高阶偏导数；如果函数的偏导数仍然可导，则偏导数的偏导被称为函数二阶偏导，例如f_{xy}，f_{yx}称为混合偏导
4. 当高阶混合偏导数在区域内连续时，求偏导的顺序不改变偏导数的结果，$f_{xy} = f_{yx}$

【解题模板】

① 根据偏导数定义代入数值计算
② 或先求出偏导函数，将除目标变量外其他变量看作常数，对目标变量求导后代入数值
③ 对已得偏导数再求对应偏导数即可得到高阶偏导数

【实战演练】

【Example1】(直接求偏导函数，将非目标变量当成常数)

Let z=arctan $\frac{x}{y}$, find z_x, z_y

【Solution】

$z_x = \frac{1}{1+(\frac{x}{y})^2}\frac{1}{y} = \frac{y}{x^2+y^2}$,

$z_y = \frac{1}{1+(\frac{x}{y})^2}(-\frac{x}{y^2}) = \frac{-x}{x^2+y^2}$

极限与连续性

【基本思想】

(1) 极限定义

To say that $\lim_{(x,y)\to(a,b)} f(x,y)=L$ means that for each ε > 0 (no matter how small) there is a corresponding δ > 0 such that |f(x,y)-L| < ε provided that 0 <|| (x, y) - (a, b) || < δ.

(2) 不同路径检验法

If f(x, y)→L_1 as (x, y)→ (a, b) along a path C_1 and f(x, y)→L_2 as (x, y)→(a, b) along a path C_2, where $L_1 \neq L_2$, then $\lim_{(x,y)\to(a,b)} f(x,y)$ does not exist.

(3) 连续定义

If $\lim_{(x,y)\to(a,b)} f(x,y)$ exists and f(a, b) is defined, $\lim_{(x,y)\to(a,b)} f(x,y)$=(a,b). We say f(x, y) is continuous at (a, b).

【解题模板】

与上学期的微积分类似，这学期的自变量变成两个及以上

①存在极限（limit）：无论从哪里都趋近于同一个值

Tips: 当题目明显需要证明极限不存在，则只需要找两个值，使得趋于的值不一样即可

②连续（continuity）：极限（的值）=定义点（的值）

【实战演练】

【Example 1】

Let $f(x,y)=\frac{x^2y}{x^4+y^2}$. Determine whether the following limit exists:

$$\lim_{(x,y)\to(0,0)} f(x,y).$$

【Solution】

Remark：这题可以判断为上面解题模板的①，可以直接找两个比较好算的直线假设为 y 来带入算出在（0，0），接着我们发现，这题长得不太友好，那我们可以在假设时再加入一个变量（后期随时变化来发现取值的不同）

Step 1 ：Substitute y=mx^2 into f(x,y):

$$f(x,mx^2)=\frac{x^2(mx^2)}{x^4+(mx^2)^2} = \frac{mx^4}{x^4+mx^4} = \frac{m}{1+m^2}$$

The limit depends on m, which is the slope of the parabola. For example:

If m=1, the limit is $\frac{1}{2}$. If m=0,the limit is 0

Step 2 : Substitute y=0:

$$f(x,0)=\frac{x^2(0)}{x^4+(0)^2}=0.$$

The limit is 0.

Since the limit depends on the path $\frac{1}{2}$ for y=x^2 and 0 for y=0, the limit does not exist.

【Example 2】

Consider the piecewise function:

f(x,y)= $\begin{cases} \frac{x^2-y^2}{x^2+y^2}, & \text{if } (x,y) \neq (0,0) \\ 0, & \text{if } (x,y) = (0,0) \end{cases}$, Is f(x,y) continuous at (0,0)?

【Solution】

Step 1: check the definition of continuity

For f(x,y) to be continuous at (0,0),the following must hold:

$$\lim_{(x,y)\to(0,0)} f(x,y)=f(0,0)=0$$

Step 2: compute the limit along different paths

♥Path 1:along y=0

Substitute y=0 into f(x,y):

$$f(x,0)= \frac{x^2-0^2}{x^2+0^2} = 1.$$

The limit is 1.

♥Path 2:along y=x

Substitute y=x into f(x,y):

$$f(x,x)= \frac{x^2-x^2}{x^2+x^2} = 0.$$

The limit is 0.

Since the limit depends on the path along y=0 and 0 along y=x, the limit does not exist. Therefore, f(x,y) is not continuous at (0,0).

全微分

【基本思想】
(1) 微分定义
F is locally linear or differentiable at (a,b) if

$$f(a + h_1, b + h_2) = f(a, b) + h_1 f_x(a, b) + h_2 f_y(a, b) + h_1\varepsilon_1(h_1, h_2) + h_2\varepsilon_2(h_1, h_2)$$

Where $\varepsilon_1(h_1, h_2) \to 0$ as $(h_1, h_2) \to 0$ and $\varepsilon_2(h_1, h_2) \to 0$ as $(h_1, h_2) \to 0$
(2)局部线性近似

$$f(x, y) \approx f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

(3)全微分（Total differential）

$$dz = f_x(x, y)dx + f_y(x, y)dy$$

【实战演练】
【Example 1】**全微分的计算** Compute Total Differentials

(a) Find the total differential dz for:
$z = x^2 + 3xy - y^3$,at point (1, 2).

(b) Find the total differential dz for:
$z = e^x \sin y$,at point (0, π/2).

【Solution】
(a) $dz = (2x + 3y)dx + (3x - 3y^2)dy$
At (1, 2),$dz = (2\times1 + 3\times2)dx + (3\times1 - 3\times2^2)dy = 8dx - 9dy$

(b) $dz=(e^0\sin\frac{\pi}{2})dx+(e^0\cos\frac{\pi}{2})dy=1\times dx+0\times dy=dx$

【Example 2】**判断可微性** Differentiability Verification

(a) For the function:

$$f(x,y) = \begin{cases} \dfrac{xy}{\sqrt{x^2+y^2}}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$$

Is f differentiable at (0, 0)? Explain.

(b)Proof (Relationship Between Partial Derivatives and Differentiability)偏导数与可微性的关系

Prove: If a function z = f(x, y) has continuous partial derivatives f_x and f_y at (a,

b), then f is differentiable at (a, b).

【Solution】

(a)For f(x,y) = $\frac{xy}{\sqrt{x^2+y^2}}$:

Step 1: Check existence of partial derivatives at (0,0):

$$f_x(0,0)=\lim_{h\to 0}\frac{f(h,0)-f(0,0)}{h}=0, f_y(0,0)=0$$

Step 2: Use definition of differentiability:

$$\lim_{(h,k)\to(0,0)}\frac{f(h,k)-[f(0,0)+f_x(0,0)h+f_y(0,0)k]}{\sqrt{h^2+k^2}}=\lim_{(h,k)\to(0,0)}\frac{hk}{h^2+k^2}$$

This limit does not exist (e.g., approaching along h=k: limit =1/2≠0).
Conclusion: f is not differentiable at (0,0).

(b) Please check the proof of Textbook 11.6 Theorem A.

【Example 3】**近似计算** Approximate Calculations

Estimate sqrt((3.02)² + (4.01)²) using total differentials.

【Solution】

Let f(x,y)=$\sqrt{x^2+y^2}$. At(x_0,y_0)=(3,4):

dz=$\frac{3}{5}$(0.02) + $\frac{4}{5}$(0.01)=0.016

⇒$\sqrt{3.02^2+4.01^2}$≈5+0.016=5.016

【Example 4】**几何应用** Geometric Applications--Tangent Plane Equation

Find the tangent plane to the surface z = x² - y² at point (1, 1, 0).

【Solution】

For z=x^2-y^2,the partial derivatives at(1,1,0) is,
f_x=2x=2, f_y=−2y=−2
Tangent plane equation,z−0=2(x−1)−2(y−1)
⇒2x−2y−z=0

方向向量与梯度

词汇：Directional Derivatives 方向导数
Gradient 梯度

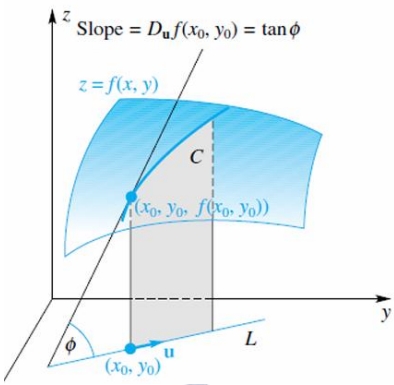
1. Directional Derivatives

【基本思想】

设函数 f 在点 P 的某个邻域内有定义，u 是单位向量. 若极限 $\lim\limits_{h\to 0}\frac{f(P+hu)-f(P)}{h}$ 存在，则其极限值称为函数 f 在点 P 处沿方向 u 的方向导数(directional derivative)，记为 $D_u f(P)=\lim\limits_{h\to 0}\frac{f(P+hu)-f(P)}{h}$ ，由定义，方向导数是在 u 方向上函数值关于距离的变化率.

【辅助理解】

在二维坐标系中，导数即为函数的变化率，同样的，在三维坐标系中，我们仍需要导数去描述函数的变化率，但不同于二维坐标系中的是，三维中函数的变化方向不再是简单的向左或向右，而是各个方向的，所以为了更好的描述函数在三维中的变化，便有了方向导数去解决问题。



【Theorem A】

若二元函数 $z=f(x,y)$ 在点 P（a，b）可微，则此函数在该点沿向量 $u(u_1 i+u_2 j)$ 有方向导数，且方向导数为$D_u f(P)=\nabla f(P)\cdot u=f_x(a,b)u_1+f_y(a,b)u_2$，其中，

$$u_1=\frac{a}{\sqrt{a^2+b^2}},\quad u_2=\frac{b}{\sqrt{a^2+b^2}}$$

2. Gradient

【基本思想】

函数 $z=f(x,y)$ 在点 P（a，b）可微，则 $\nabla f(x,y)=f_x(P)i+f_y(P)j$ 称为 $f(x,y)$ 在点 P（a，b）的梯度

【Theorem B】

若二元函数 $z=f(x,y)$ 在点 P（a，b）可微，则在该点沿方向向量 $u(u_1 i+u_2 j)$ 的方向导数与梯度的关系为 $\nabla f(x,y)\cdot u=f_x(a,b)u_1+f_y(a,b)u_2$. 另外，若用

θ表示在点 $P(x_0, y_0)$ 梯度与单位向量 u 的夹角，则

$$D_u f(P) = \|\nabla f(P)\| \cdot \|u\| \cdot \cos\theta = \sqrt{f_x^2(x_0, y_0) + f_y^2(x_0, y_0)} \cos\theta$$

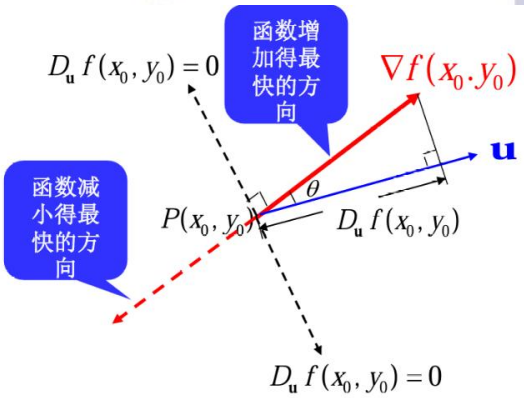
当$\theta = 0$ 方向导数$D_u f(P)$取得最大值

当$\theta = \pi$ 方向导数$D_u f(P)$取得最小值

当$\theta = \frac{\pi}{2}$或$\frac{3\pi}{2}$，方向导数$D_u f(P) = 0$

【Theorem C】

函数在一点处的梯度的方向是函数在该点取得最大方向导数的方向（即函数在该点增加最快的方向，$\|\nabla f(P)\|$ 是函数在该点的方向导数的最大值，梯度方向的反方向是函数在该点取得最小方向导数的方向（即函数在该点减少最快的方向），此时方向导数为 $-\|\nabla f(P)\|$



【实战演练】

【Example 1】

Find the gradient vector of the given function at the given point p. Then find the equation of the tangent plane at p（-2，3）

$$f(x, y) = x^2y - xy^2,\ p = (-2,3)$$

【Solution】

$$\nabla f(x, y) = \langle x(x, y), f_y(x, y)\rangle = \langle 2xy - y^2, x^2 - 2xy\rangle$$

$$\nabla f(-2,\ 3) = \langle -21,16\rangle$$

The equation of the tangent plane at p（-2，3）

$z = f(-2,3) - 21(x + 2) + 16(y - 3) = -21x + 16y - 60$

【Example 2】

Find the directional derivative of $f(x,y) = e^x \cdot \sin y$ at the point p = $\left(0, \frac{\pi}{4}\right)$, in the direction of a = i + $\sqrt{3}$i

【Solution】

$\nabla f(x,y) = \left\langle f_\chi(x,y), f_y(x,y) \right\rangle = \langle e^x \sin y, e^x \cos y \rangle$

$$\nabla f\left(0, \ \frac{\pi}{4}\right) = \left\langle \frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2} \right\rangle$$

The unit vector in the direction of a = i + $\sqrt{3}$i is

$$u = \frac{a}{\|a\|} = \frac{1}{2} \ (i + \sqrt{3}i) = \left\langle \frac{1}{2}, \frac{\sqrt{3}}{2} \right\rangle$$

Therefore

$$D_u f\left(0, \frac{\pi}{4}\right) = \nabla f\left(0, \frac{\pi}{4}\right) = \frac{1}{4}\left(\sqrt{2} + \sqrt{6}\right)$$

多元函数求导

专题数学词汇： partial derivatives 偏导数
The chain rule 链式法则
implicit function 隐函数

【基本思想】

1. Partial Derivatives 偏导数：

Suppose that f is a function of two variables x and y. If y is held constant, say $y = y_0$, then $f(x, y_0)$ is a function of the single variable x. Its derivative at $x = x_0$ is called the partial derivative of f with respect to x at (x_0, y_0) and is denoted by $\mathbf{f_x}(x_0, y_0)$.

$$\mathbf{f_x}(x_0, y_0) = \lim_{\Delta x \to 0} \frac{f(x_0 + \Delta x, y_0) - f(x_0, y_0)}{\Delta x}$$

【辅助理解】:

① 由二元偏导的定义，我们还可以推出三元偏导的定义：

$$\mathbf{f_x}(x_0, y_0, z_0) = \lim_{\Delta x \to 0} \frac{f(x_0 + \Delta x, y_0, z_0) - f(x_0, y_0, z_0)}{\Delta x}$$

②当高阶混合偏导数在区域内连续时，求偏导的顺序不改变偏导数的结果

2. The Chain Rule 链式法则：

Let $x = x(t)$ and $y = y(t)$ be differentiable at t, and let $z = f(x, y)$ be differentiable at $(x(t), y(t))$. Then $z = f(x(t), y(t))$ is differentiable at t and

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$$

【辅助理解】: 类似于一元微积分的链式法则

【解题模板】

1. 多元函数偏导：对于某一变量求偏导，把其他元素当作常量。
2. 隐函数：应用隐函数定理

$$\frac{dy}{dx} = -\frac{F_x(x, y)}{F_y(x, y)}$$

3. 链式法则：寻找中间变量和独立变量

【实战演练】

【Example1】（链式法则）

$$\text{The function } z = z(x,y) \text{ satisfies } x^2\frac{\partial z}{\partial x} + y^2\frac{\partial z}{\partial y} = z^2 .$$

$$\text{Suppose } u = x\,,\, v = \frac{1}{y} - \frac{1}{x}, \varphi = \frac{1}{z} - \frac{1}{x} \text{ . To the function } \varphi$$

$$= \varphi(u,v)\text{ , prove that: } \frac{\partial \varphi}{\partial u} = 0$$

【Tips】

Using the chain rule, we can find our solution direction:

$$\frac{\partial \varphi}{\partial u} = \frac{\partial \varphi}{\partial x}\cdot\frac{\partial x}{\partial u} + \frac{\partial \varphi}{\partial y}\cdot\frac{\partial y}{\partial u}$$

【Solution1】（链式法则）

Simply analyzing the question reveals that:

$$\begin{cases} u = x \\ v = \dfrac{1}{y} - \dfrac{1}{x} \end{cases} \Rightarrow \begin{cases} x = u \\ y = \dfrac{u}{uv+1} \end{cases}$$

And

$$\frac{\partial x}{\partial u} = 1\,;\ \frac{\partial y}{\partial u} = \frac{(uv+1) - u\cdot v}{(uv+1)^2} = \frac{1}{(uv+1)^2}$$

Besides,

$$\varphi = \frac{1}{z} - \frac{1}{x} = \frac{1}{z(x,y)} - \frac{1}{x}$$

$$\therefore\ \frac{\partial \varphi}{\partial x} = \frac{-1}{z^2}\cdot\frac{\partial z}{\partial x} + \frac{1}{x^2}\quad ;\quad \frac{\partial \varphi}{\partial y} = -\frac{1}{z^2}\cdot\frac{\partial z}{\partial y}$$

Thus, bringing into the equation above:

$$\frac{\partial \varphi}{\partial u} = \left(-\frac{1}{z^2}\cdot\frac{\partial z}{\partial x} + \frac{1}{x^2}\right) + \left(-\frac{1}{z^2}\cdot\frac{\partial z}{\partial y}\right)\cdot\frac{1}{(uv+1)^2} = -\frac{1}{z^2}\frac{\partial z}{\partial x} + \frac{1}{x^2} - \frac{1}{z^2}\cdot\frac{\partial z}{\partial y}\cdot\frac{y^2}{x^2}$$

$$= -\frac{1}{z^2}\left(\frac{1}{x^2}\cdot\left[x^2\cdot\frac{\partial z}{\partial x} + y^2\cdot\frac{\partial z}{\partial y}\right]\right) + \frac{1}{x^2} = -\frac{1}{x^2} + \frac{1}{x^2} = 0$$

【Solution2】(Complete differential 全微分)

We also usually use another way which presents all provided equations' differential to solve it:

$$\begin{cases} dz = \frac{\partial z}{\partial x} \cdot dx + \frac{\partial z}{\partial y} \cdot dy \ (1) \\ du = dx \ \ (2) \\ dv = -\frac{1}{y^2} \cdot dy + \frac{1}{x^2} \cdot dx \ (3) \\ dy = -\frac{1}{z^2} dz + \frac{1}{x^2} \cdot dx \ \ (4) \end{cases}$$

Combine the above four equations:

$$\frac{\partial \varphi}{\partial u} = 0$$

【Example 2】（隐函数）

$$\text{If } \cos^2 x + \cos^2 y + \cos^2 z = 1 \quad , dz =$$

【Solution】

$$dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy$$

Using implicit function theorem:

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z} = -\frac{\cos x \sin x}{\cos z \sin z} = -\frac{\sin 2x}{\sin 2z}$$

$$\frac{\partial z}{\partial y} = -\frac{F_y}{F_z} = -\frac{\cos y \sin y}{\cos z \sin z} = -\frac{\sin 2y}{\sin 2z}$$

Thus, $dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy = -\frac{1}{\sin 2z}(\sin 2x \, dx + \sin 2y \, dy)$

【Example 3】（高阶复合偏导）

If $3\frac{\partial^2 z}{\partial x^2} - 2 \cdot \frac{\partial^2 z}{\partial x \partial y} - \frac{\partial^2 z}{\partial y^2} = 0$ and $\begin{cases} u = x + ay \\ v = x - 3y \end{cases}$, they can combine to prove $\frac{\partial z}{\partial u \partial v} = 0$. Find the value of a

【Solution】

Simply analyzing the question reveals that:

$$\begin{cases} x = u - ay = u - \frac{a(u - v)}{a + 3} = \frac{3u + av}{a + 3} \\ y = \frac{u - v}{a + 3} \end{cases}$$

$$\begin{cases} \dfrac{\partial z}{\partial u}=\dfrac{\partial z}{\partial x}\cdot\dfrac{\partial x}{\partial u}+\dfrac{\partial z}{\partial y}\cdot\dfrac{\partial y}{\partial u}=\dfrac{\partial z}{\partial x}\cdot\dfrac{3}{a+3}+\dfrac{\partial z}{\partial y}\cdot\dfrac{1}{a+3} \\ \dfrac{\partial z}{\partial v}=\dfrac{\partial z}{\partial x}\cdot\dfrac{\partial x}{\partial v}+\dfrac{\partial z}{\partial y}\cdot\dfrac{\partial y}{\partial v}=\dfrac{\partial z}{\partial x}\cdot\dfrac{a}{a+3}+\dfrac{\partial z}{\partial y}\cdot\left(-\dfrac{1}{a+3}\right) \end{cases}$$

And,

$$\frac{\partial z}{\partial u\partial v}=\frac{\partial\left(\frac{\partial z}{\partial u}\right)}{\partial v}=\frac{\partial\left(\frac{\partial z}{\partial u}\right)}{\partial x}\frac{\partial x}{\partial v}+\frac{\partial\left(\frac{\partial z}{\partial u}\right)}{\partial y}\cdot\frac{\partial y}{\partial v}$$

$$=\left(\frac{\partial^2 z}{\partial x^2}\cdot\frac{3}{a+3}+\frac{\partial^2 z}{\partial x\partial y}\cdot\frac{1}{a+3}\right)\cdot\frac{a}{a+3}+\left(\frac{\partial^2 z}{\partial x\partial y}\cdot\frac{3}{a+3}+\frac{\partial^2 z}{\partial y^2}\cdot\frac{1}{a+3}\right)\cdot\left(-\frac{1}{a+3}\right)=0$$

$$\text{Since}\quad 3\frac{\partial^2 z}{\partial x^2}-2\cdot\frac{\partial^2 z}{\partial x\partial y}-\frac{\partial^2 z}{\partial y^2}=0$$

$$\frac{\partial^2 z}{\partial x^2}\cdot\frac{3a}{(a+3)^2}+\frac{\partial^2 z}{\partial x\partial y}\cdot\frac{a-3}{(a+3)^2}+\frac{\partial^2 z}{\partial y^2}\cdot\left(-\frac{1}{(a+3)^2}\right)=0$$

$$\therefore\ \frac{3a}{(a+3)^2}:\frac{a-3}{(a+3)^2}:-\frac{1}{(a+3)^2}=3:-2:-1$$

$$\Rightarrow\ \frac{a-3}{(a+3)^2}/-\frac{1}{(a+3)^2}=-2/-1$$

$$\therefore\quad 3-a=2$$

$$\therefore\quad a=1$$

多元函数求极值

【基本思想】

Pay attention to

1. Stationary Points 驻点
   (where the gradient is zero).
2. Singular Points 奇异点
   (where the function is non-differentiable or discontinuous).
3. Boundary Points 边界点
   (if the domain is restricted, extrema may occur on the boundary).

【解题模板】

1. Find Stationary Points
   ①Compute the gradient ∇f
   ②Solve $\nabla f=0$ for stationary points.
   ③Use the Hessian matrix to classify them：

   $D = f_{xx}f_{yy}-(f_{xy})^2$：

   - $D>0$ and $f_{xx}>0$：local minimum value；
   - $D>0$ and $f_{xx}<0$：local maximum value；
   - $D<0$：not an extreme value，where is a saddle point；
   - $D=0$：inconclusive

2. Find Singular points
   Identify non-differentiable regions: Check where partial derivatives do not exist.
   Evaluate f at singular points

3. Find Boundary Points
   Parametrize the boundary
   Find the value

【实战演练】

【Example 1】

Find extreme value of $f(x,y) = x^3+ y^3- 3x - 3y$：

【Solution】

Let $f_x=3x^2-3 = 0$

$f_y=3y^2-3 = 0$

Have the stationary points(±1,±1)

And : $D = f_{xx}f_{yy}-(f_{xy})^2 = 36xy$

At (1,1): D=36×1×1=36 and f_{xx}=6 , local minimum.
At (1,−1): D=36×1×(−1)=−36 , so this is a saddle point.
At (−1,1): D=36×(−1)×1=−36 , so this is a saddle point.
At (−1,−1): D=36×(−1)×(−1)=36 , f_{xx}=−6 ,local maximum.
f(1,1) =−4 , f(−1,−1)= =4
Thus, the function has a local minimum of -4 at (1,1) and a local maximum of 4 at (−1,−1)

【辅助理解】
1. 求梯度：对 x，y 分别求偏导
2. 找驻点：令偏导等于 0 得驻点 (±1,±1)
3. 计算 Hessian 矩阵：D = $f_{xx}f_{yy}-(f_{xy})^2$ = 36xy
4. 判断极值

【Example 2】

Find the maximum and minimum values of (x,y) =2 + x^2+ y^2 on the closed and bounded set S = {(x,y) : $x^2 + \frac{y^2}{4} \le 1$ }

【Solution】

Let f_x=2x= 0 , f_y=2y = 0, x=0,y=0

f(0,0)=2+0+0=2
Parametrize the boundary
x=cosθ , y=2sinθ (θ∈[0,2π)).
Then
f(cosθ,2sinθ) = 2 + $\cos^2\theta + (2\sin\theta)^2$ = 2 + $\cos^2\theta + 4\sin^2\theta$
f = 2 + (1−$\sin^2\theta$) + 4$\sin^2\theta$ = 3 +3 $\sin^2\theta$
Since
$\sin^2\theta \in [0,1]$，f ∈ [3,6].
When sinθ = 0 ， f=3
When sinθ=±1 ， f=6
And 3 < f(0,0)=2
Hence the minimum value is 2 at (0,0).
the maximum value is 6 at (0,2) and (0,−2)

【辅助理解】
1. 寻找内部驻点（0，0）
2. 通过参数化分析边界极值
   f(cosθ, 2sinθ) = 2 + $\cos^2\theta + (2\sin\theta)^2$ = 2 + $\cos^2\theta + 4\sin^2\theta$
3. 对比分析，下结论

柱面及球面坐标系求体积

【基本思想】

1. 柱面坐标系：

将直角坐标系中的点用极坐标（二维平面）和高度（z 轴）表示

设参：

r：点到 z 轴的径向距离，$r \geq 0$

θ：绕 z 轴的旋转角度，$0 \leq \theta < 2\pi$

z：沿 z 轴的高度（与直角坐标系相同）

适用场景：圆柱形、旋转对称问题

2. 球面坐标系

将直角坐标系中的点用点到原点的距离和两个角度表示

设参：

ρ：点到原点的距离，$\rho \geq 0$

θ：绕 z 轴的方位角，$0 \leq \theta < 2\pi$

ϕ：点与 z 轴的极角，$0 \leq \phi \leq \pi$

适用场景：球体、圆锥、球壳等球对称问题

【解题模板】

①依据题意将直角坐标系转换为柱面或球面坐标

柱面坐标 ：$\begin{cases} x = r\cos\theta \\ y = r\sin\theta \\ z = z \end{cases}$

球面坐标 ：$\begin{cases} x = \rho\sin\phi\cos\theta \\ y = \rho\sin\phi\sin\theta \\ z = \rho\cos\phi \end{cases}$

②确定积分参数范围

柱面坐标：r（半径范围）、θ（角度范围）、z（高度范围）

球面坐标：ρ（半径范围）、ϕ（极角范围）、θ（方位角范围）

③积分 dV

柱面坐标 ：$dV = r\,dr\,d\theta\,dz$

球面坐标 ：$dV = \rho^2 \sin\phi\,d\rho\,d\phi\,d\theta$

④总结如下表：

	直角坐标系转换公式	dV
柱面坐标	$\begin{cases} x = r\cos\theta \\ y = r\sin\theta \\ z = z \end{cases}$	$dV = r\,dr\,d\theta\,dz$
球面坐标	$\begin{cases} x = \rho\sin\phi\cos\theta \\ y = \rho\sin\phi\sin\theta \\ z = \rho\cos\phi \end{cases}$	$dV = \rho^2 \sin\phi\,d\rho\,d\phi\,d\theta$

【实战演练】
【Example 1】
Compute the volume of a cylinder $x^2+y^2\le 4$，$0\le z\le 3$

【Solution】
In cylindrical coordinates
let $r \in [0,2]$, $\theta \in [0,2\pi)$, $z \in [0,3]$

Then V= $\int_0^{2\pi}\int_0^2\int_0^3 r\,dz\,dr\,d\theta = 2\pi \int_0^2 3r dr = 12\pi$.

【辅助理解】
① 求圆柱体体积，选择柱面坐标系
② 得到柱面坐标系下各参数范围 $r \in [0,2]$, $\theta \in [0,2\pi)$, $z \in [0,3]$
③ 体积积分 V= $\int_0^{2\pi}\int_0^2\int_0^3 r\,dz\,dr\,d\theta$

【Example 2】
Find the volume of a sphere $x^2+ y^2+ z^2\le 9$

【Solution】
In spherical coordinates:
let $\rho \in [0,3]$, $\phi \in [0,\pi]$, $\theta \in [0,2\pi)$

Then V= $\int_0^{2\pi}\int_0^{\pi}\int_0^3 \rho^2 \sin\phi\, d\rho\, d\phi\, d\theta = 36\pi$.

【辅助理解】
求球体积，选择球面坐标系

【Example 3】

Calculate the volume bounded by z= $\sqrt{x^2+y^2}$ and z = 1

【Solution】
In cylindrical coordinates
let $z \in [r,1]$, $r \in [0,1]$ $\theta \in [0,2\pi)$

Then V= $\int_0^{2\pi}\int_0^1\int_r^1 r\,dz\,dr\,d\theta = \frac{\pi}{3}$

【辅助理解】
有圆锥和平面均绕 z 轴对称，选择柱面坐标系使得计算简便

二重积分计算（一般情况）

数学词汇：Multiple Integrals 多重积分
Double Integral 二重积分

【基本思想】

Let f be a function of two variables that is defined on a closed rectangle R. If . $\lim_{\|P\|\to 0}\sum_{k=1}^{n} f(\bar{x}_k,\bar{y}_k)\Delta A_k$ exists, we say that f is integrable on R. Moreover, $\iint_R f(x,y)dA$, called the double integral of f over R, is then given by $\iint_R f(x,y)dA = \lim_{\|P\|\to 0}\sum_{k=1}^{n} f(\bar{x}_k,\bar{y}_k)\Delta A_k$ P, denoted by ‖P‖, is the length of the longest diagonal of any subrectangle in the partition.

【Theorem A】

If f(x , y) is continuous throughout the rectangular region R: a ≤ x ≤ b, c ≤ y ≤ d,

then $\iint_R f(x,y)dA = \int_c^d\int_a^b f(x,y)dxdy = \int_a^b\int_c^d f(x,y)dy\,dx$

【Theorem B】

比较定理：在 R 上,f(x,y)<g(x,y), $\iint_R f(x,y)dA < \iint_R g(x,y)dA$

【Theorem C】

Sum and Difference

$\iint_R [f(x,y) \pm g(x,y)]dA = \iint_R f(x,y)dA \pm \iint_R g(x,y)dA$

【Theorem D】

If R is the union of two nonoverlapping regions R_1 and R_2 then

$\iint_R f(x,y)dA = \iint_{R_1} f(x,y)dA + \iint_{R_2} f(x,y)dA$

【确定积分限的解题步骤】

1. 首先确定是先对 x 还是 y 积分
2. 画图. 画出积分区域的草图并标示出边界曲线.
3. 假如先对 x 后对 y 积分，先**确定 y 的范围**：找到区域 D 在 y 轴上的投影区间 [c ， d]，即 y 的下限为 c，上限为 d
4. **确定 x 的范围**：对任意固定的 y∈[c，d]，作水平于 y 轴的直线穿过区域 D，该直线与区域边界的交点的 x 坐标即为 x 的下限和上限。通常：下限为左侧边界曲线 $x=x_1$ （y），上限为右侧边界曲线 $x=x_2$ （y）

5. 最终积分限表示为：$\iint_D f(x,y)dxdy = \int_c^d \left[\int_{x_1(y)}^{x_2(y)} f(x,y)dx\right] dy$

【实战演练】
【Example】

Compute：$\iint_D f(x,y)dxdy$，where D is enclosed by y = x, y = x^2

【Solution】

Step 1: Determine the Intersection Points of the Boundaries

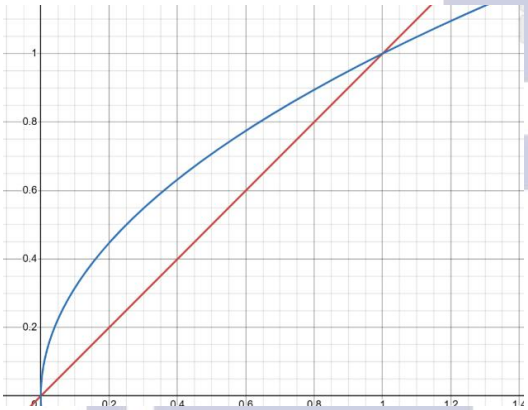
Substitute y=x into y = x^2

The corresponding intersection points are:

(0,0) when x=0,

(1,1) when x=1.

Step 2: Analyze the Region D



Step 3: Set Up the Double Integral

the integral is structured as: $\int_0^1 \int_{x^2}^{x} xydy\, dx$

Step 4: Evaluate inner and outer

$$\int_{x^2}^{x} xydy = \frac{1}{2}(x^3 - x^5)$$

$$\frac{1}{2}\int_0^1 (x^3 - x^5)dx = \frac{1}{24}$$

Thus，the answer is $\frac{1}{24}$

极坐标二重积分

专题数学词汇： Polar coordinates 极坐标
Double integrals 二重积分

【基本思想】

3. Double Integrals in Polar Coordinates 极坐标的二重积分：

Let R have the shape shown at the left in the figure below, which we call a polar rectangle and will describe analytically in a moment. Let z = f (x, y) determine a surface over R and suppose that f is continuous and nonnegative. Then the volume V of the solid under this surface and above R (as shown in the middle in the figure below) is given by

$$\mathbf{V} = \iint_{\mathbf{R}} \mathbf{f}(\mathbf{x}, \mathbf{y}) \mathrm{d}\mathbf{A}$$

In polar coordinates, a polar rectangle R has the form:

$$\mathrm{R} = \{(\mathrm{r}, \theta): \mathrm{a} \leq \mathrm{r} \leq \mathrm{b}, 2 \leq \theta \leq 13\}$$

where α > 0 and β − α ≤ 2π. Also, the equation of the surface can be written as:

$$\mathrm{z} = \mathrm{f}(\mathrm{x}, \mathrm{y}) = \mathrm{f}(\mathrm{r}\cos\theta, \mathrm{r}\sin\theta) = \mathrm{F}(\mathrm{r}, \theta)$$

We are going to calculate the volume V in a new way using polar coordinates. Partition R into smaller polar rectangles R1, R2, ..., Rn by means of a polar grid and let Δr_k and $\Delta\theta_k$ denote the dimensions of the typical piece Rk, as shown at the right in the figure below. The area A(Rk) is given by:

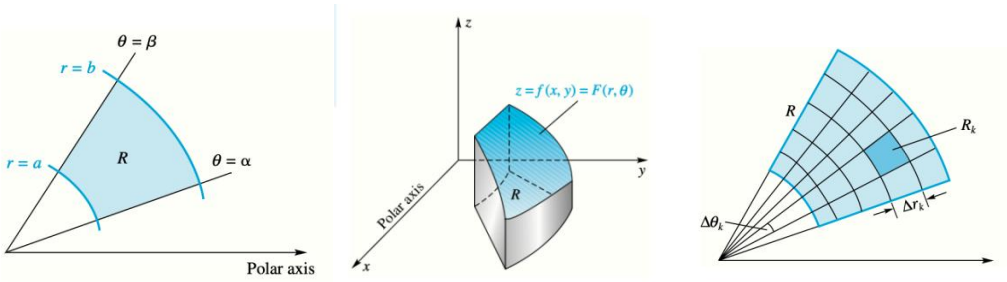
$$\mathrm{A}(\mathrm{R}_k) = \overline{\mathrm{r}_k}\, \Delta \mathrm{r}_k\, \Delta\theta_k$$

where $\overline{\mathrm{r}_k}$ is the average radius of Rk. Thus,

$$\mathrm{V} \approx \sum_{k=1}^{n} 1 = (\overline{\mathrm{r}_k}, \overline{\theta}_k)\overline{\mathrm{r}_k}\, \Delta\, \mathrm{r}_k\, \Delta\, \theta_k$$

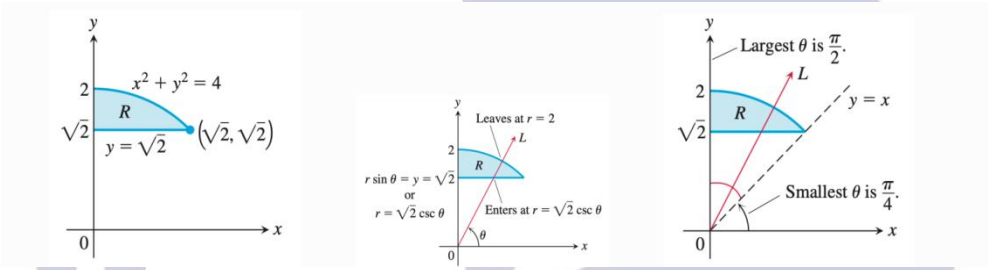
When we take the limit as the norm of the partition approaches zero, we ought to get the actual volume. This limit is a double integral.

$$\mathrm{V} = \iint_{\mathrm{R}} \mathrm{F}(\mathrm{r}, \theta)\mathrm{r}\, \mathrm{d}\mathrm{r}\mathrm{d}\theta = \iint_{\mathrm{R}} \mathrm{f}(\mathrm{r}\cos\theta, \mathrm{r}\sin\theta)\, \mathrm{r}\, \mathrm{d}\mathrm{r}\mathrm{d}\theta$$



【解题模板】

4. 画草图. 画出区域。
5. 找 r 的积分限. 设想一射线 L 从原点出发以 r 增加的方式的穿过 R. 标示出射线 L 穿入和穿出的 r 值, 它们就是 r 的积分限, 且通常依赖角 θ, θ 为 L 与 x 轴正向的夹角.
6. 找 θ 的积分限. 找出 R 边界的最小和最大的 θ 值, 它们是 θ 的积分限.



【实战演练】

【Example1】

Find the value of:

$$\iint_D \sqrt{x^2+y^2}\,dxdy \quad , \quad D:\{(x,y) \mid x^2+y^2 \le 2x, 0 \le y \le x\}$$

【Solution】

Simply analyze the information:

$$x^2-2x+y^2 \le 0 \Rightarrow (x-1)^2+y^2 \le 1^2 \text{ and } 0 \le y \le x$$

The area of D is

$$\iint_D \sqrt{x^2+y^2}\,dxdy = \iint_D r \cdot r\,dr\,d\theta = \int_0^{\frac{\pi}{4}} d\theta \int_0^{2\cos\theta} r^2 dr = \int_0^{\frac{\pi}{4}} \frac{1}{3} r^3 \Big|_0^{2\cos\theta} d\theta$$

$$= \frac{8}{3}\int_0^{\frac{\pi}{4}} \cos^3\theta\, d\theta = \frac{8}{3}\left(\sin\theta - \frac{1}{3}\sin^3\theta \Big|_0^{\frac{\pi}{4}}\right) = \frac{10}{9}\sqrt{2}$$

【Example2】

Find the value of:

$$\iint\limits_{x^2+y^2\le 1} (2x+3y)^2 d\sigma$$

【Solution】

$$\iint\limits_{x^2+y^2\le 1} (2x+3y)^2 d\sigma = \iint\limits_{x^2+y^2\le 1} 4x^2+12xy+9y^2 d\sigma$$

Due to 12xy is an odd function, then we can notice that:

$$\iint\limits_{x^2+y^2\le 1} 12xy d\sigma = 0$$

Thus,

$$\iint\limits_{x^2+y^2\le 1} (2x+3y)^2 d\sigma = \iint\limits_{x^2+y^2\le 1} 4x^2+9y^2 d\sigma$$

Because the area of $x^2+y^2 \le 1$ takes turns the symmetry,

$$\iint\limits_{x^2+y^2\le 1} 4x^2+9y^2 d\sigma = \frac{13}{2} \iint\limits_{x^2+y^2\le 1} (x^2+y^2) d\sigma = \frac{13}{2}\int_0^{2\pi} d\theta \int_0^1 r^3 dr = \frac{13}{4}\pi$$

曲面面积

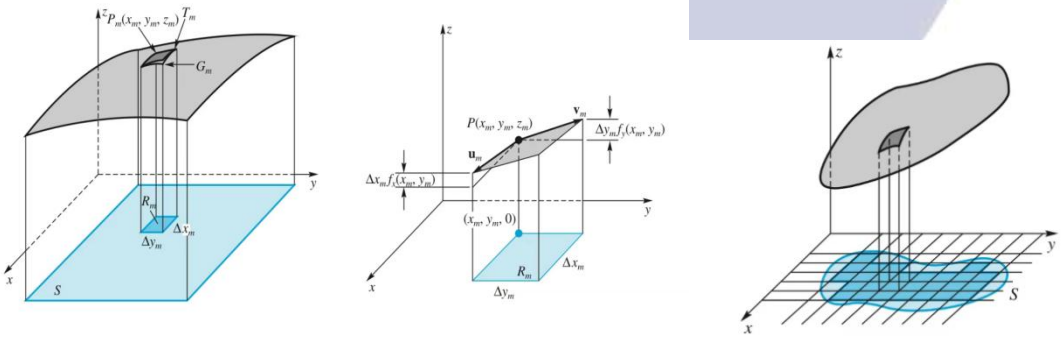
专题数学词汇：centroid 质心
circumference 圆周周长
closed and bounded region 有界闭区域
hemisphere 半球
horizontal plane 水平平面
hyperbolic paraboloid 双曲抛物面
inscribed sphere 内切的球
parallelogram 平行四边形
perpendicular to… 与… 垂直
planar region 平面区域
project onto… 投射到…上
projection 投影
right circular cylinder 直圆柱
surface area of a sphere 球面面积
tangent parallelogram 切平行四边形

【基本思想】

1，用二重积分计算曲面表面积的核心：
将曲面分割为无数微小的切平面（或近似平面），计算每个微小区域的面积，再通过二重积分求和。

2，曲面 S 在 xy-平面的投影为 D（积分区域）

3，曲面面积公式推导：
根据图片，在点 P 分别写出其在 x 轴和 y 轴移动Δx 和Δy 个单位长度的切线方向向量，通过计算两个切向量叉乘的模求出其微小切平面的面积，最后积分求和即得出曲面表面积的大小。



$$\mathbf{u}_m = \Delta x_m \mathbf{i} + f_x(x_m + y_m)\Delta x_m \mathbf{k}$$

$$\mathbf{v}_m = \Delta y_m \mathbf{j} + f_y(x_m + y_m)\Delta y_m \mathbf{k}$$

$$A(T_m) = \|\mathbf{u}_m \times \mathbf{v}_m\| = A(R_m)\sqrt{[f_x(x_m, y_m)]^2 + [f_y(x_m, y_m)]^2 + 1}$$

$$A(G) = \lim_{\|p\|\to 0} \sum_{m}^{1} A(T_m) = \iint_S \sqrt{[f_x(x_m, y_m)]^2 + [f_y(x_m, y_m)]^2 + 1}\ dA$$

4, 理解与拓展：

设曲面方程 $z = f(x, y)$, 其切平面的法向量为 $n = \langle f_x, f_y, -1\rangle$, 切平面的面积微元$\Delta S$ 与投影微元ΔA 的关系由法向量的模决定，$\Delta S \approx \sqrt{1 + {f_x}^2 + {f_y}^2}\Delta A$，其中$\sqrt{1 + {f_x}^2 + {f_y}^2}$是面积伸缩因子（雅可比行列式），曲面面积公式也可通过参数化的雅可比矩阵来推导（该矩阵的列向量即是参数方向的切向量，再求两个切向量叉乘后的模长即可得出）

【解题模板】

① 确定函数 $z = f(x, y)$和积分区域

② 求出两个偏导数$\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}$

③ 套用公式

$$\iint_S \sqrt{{f_x}^2 + {f_y}^2 + 1}\, dA$$

【实战演练】

【Example 1】

The part of the surface $x^2 + y^2 + z^2 = a^2$ inside the circular cylinder $x^2 + y^2 = b^2$, where $0 < b \le a$

【Solution】

$$\frac{\partial z}{\partial x} = \frac{-x}{\sqrt{a^2 - x^2 - y^2}}, \frac{\partial z}{\partial y} = \frac{-y}{\sqrt{a^2 - x^2 - y^2}}$$

Let $x = r\cos\theta, y = r\sin\theta$, the integral region is a circle of radius b.
You need to notice that the volume is divided into two halves to calculate in the case. (Not the only method)

$$S = 2\iint_D \frac{a}{\sqrt{a^2 - y^2 - x^2}} dxdy = 2\int_0^{2\pi}\int_0^b \frac{a}{\sqrt{a^2 - r^2}} rdrd\theta$$

$$= \int_0^{2\pi}\int_0^b \frac{-a}{\sqrt{a^2 - r^2}} d(a^2 - r^2)d\theta = 4\pi a(a - \sqrt{a^2 - b^2})$$

【Example 2】

Calculate that the area of the surface G cut from $z = x^2 + y^2$, by the plane $z = 0$ and $z = 1$.

【Solution】

$$A = \iint_D \sqrt{1 + 4x^2 + 4y^2}\,dA$$

Let $x = r\cos\theta, y = r\sin\theta$, the integral region is a circle of radius 1. (Sometimes it may be a circular ring, for example between $z = 1$ and $z = 2$)

$$A = \int_0^{2\pi}\int_0^1 \sqrt{1 + 4r^2}\,r\,dr\,d\theta = \int_0^{2\pi}\int_0^1 \frac{1}{8}\sqrt{1 + 4r^2}\,d(1 + 4r^2)\,d\theta$$

Next, let $u = 1 + 4r^2$, then $u = 5$ when $r = 1$ and $u = 1$ when $r = 0$

$$A = \int_0^{2\pi}\int_1^5 \sqrt{u}\,du\,d\theta = \frac{5\sqrt{5} - 1}{6}\pi$$

交换积分顺序

【基本思想】

1. 画图→找积分区域
2. 若从 dydx 变为 dxdy，就找对于每一个 y，x 从哪积到哪(注意有可能要分段看)，然后再找 y 的范围；若从 dxdy 变为 dydx，就找对于每一个 x，y 从哪积到哪(注意有可能要分段看)，然后再找 x 的范围.

【实战演练】

【Example 1】

Evaluate $\int_0^{\ln 2}\int_{e^y}^{2}\frac{e^{xy}}{x^x-1}dxdy$.

【辅助理解】

$\frac{e^{xy}}{x^x-1}$没办法直接对 x 积分，所以要交换积分次序

对于每一个 y，x 从e^y积到 2, $0\le y\le \ln 2$

$x = e^y \to y=\ln x$

画出积分区域，发现对于每一个 x，y 从 0 积到 lnx，$1\le x\le 2$

【Solution】

$\int_0^{\ln 2}\int_{e^y}^{2}\frac{e^{xy}}{x^x-1}dxdy$

$=\int_1^2\int_0^{\ln x}\frac{e^{xy}}{x^x-1}dydx$

$=\int_1^2\frac{1}{x^x-1}\cdot[\frac{e^{xy}}{x}]_0^{\ln x}dx$

$=\int_1^2\frac{1}{x^x-1}\cdot\frac{1}{x}\cdot(e^{x\ln x}-1)dx$

$=\int_1^2\frac{1}{x}dx$

$=[\ln|x|]_1^2$

$=\ln 2$

$\frac{e^{xy}}{x^x-1}$没办法直接对 x 积分，所以要交换积分次序

对于每一个 y，x 从e^y积到 2, $0\le y\le \ln 2$

$x = e^y \to y=\ln x$

画出积分区域，发现对于每一个 x，y 从 0 积到 lnx，$1\le x\le 2$

$$e^{x\ln x}-1=e^{\ln x^x}-1=x^x-1$$

【Example 2】

Write the given iterated integral at an iterated integral with the order of integration interchanged

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{2}}\int_0^{2\cos\theta}f(r\cos\theta,r\sin\theta)rdrd\theta$$

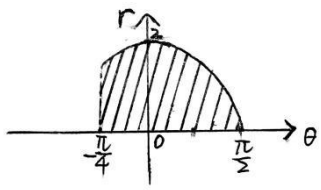
.

【Solution】

可以把 r ,θ分别看成 y， x， 即先 r 后θ →先 y 后 x

r= 2cosθ

$\theta = \arccos\frac{r}{2}$



由画图得 r∈ $[0,\sqrt{2}]$时，对于每一个 r， θ从$-\frac{\pi}{4}$ 积到 $\arccos\frac{r}{2}$

r∈ $[\sqrt{2},2]$时，对于每一个 r， θ从$-\arccos\frac{r}{2}$ 积到 $\arccos\frac{r}{2}$

$$\int_0^{\sqrt{2}}\int_{-\frac{\pi}{4}}^{\arccos\frac{r}{2}} f(r\cos\theta, r\sin\theta)r\,d\theta\,dr = \int_{\sqrt{2}}^{2}\int_{-\arccos\frac{r}{2}}^{\arccos\frac{r}{2}} f(r\cos\theta, r\sin\theta)r\,d\theta\,dr$$

【Example 3】

Evaluate $\int_0^1\int_0^{1-x}\int_{x+y}^{1}\frac{\sin z}{z}dz\,dy\,dx$.

【Solution】

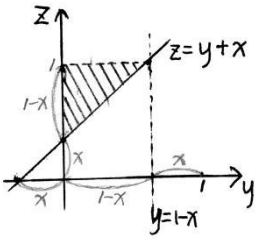
$\frac{\sin z}{z}$ 没有初等原函数，没办法对 z 积分，所以要换积分次序，变成先对 x 积分或先对 y 积分，但是 z 和 x 之间隔着 y， 不方便交换 z 和 x，所以先换 z 和 y， 对 y 积分之后再换 z 和 x

在换 z 和 y 积分顺序时，把 x 看成一个 0 到 1 之间的常数

$\int_0^{1-x}\int_{x+y}^{1}\frac{\sin z}{z}dz\,dy$

$= \int_x^1\int_0^{z-x}\frac{\sin z}{z}dy\,dz$

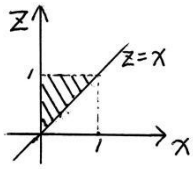
$= \int_x^1\frac{\sin z}{z}(z-x)dz$



再换 z 和 x

$\int_0^1\int_x^1\frac{\sin z}{z}(z-x)dz\,dx$

$= \int_0^1\int_0^z\frac{\sin z}{z}(z-x)dx\,dz$



$= \int_0^1\left[x\sin z - \frac{x^2}{2}\cdot\frac{\sin z}{z}\right]_0^z dz$

$= \frac{1}{2}\int_0^1 z\sin z\,dz$

$= -\frac{1}{2}\int_0^1 z\,d\cos z$

分部积分法

$= -\frac{1}{2}\left([z\cos z]_0^1 - \int_0^1\cos z\,dz\right)$

$= \frac{\sin 1-\cos 1}{2}$

无穷数列

【基本思想】

1. 数列定义

A sequence is an ordered list of numbers arranged in a row:$a_1, a_2, \ldots, a_n, \ldots$

Or, a function defined on positive integers:$a_n = f(n) \quad (n = 1,2,\ldots)$

Notation:$\{a_n\}_{n=1}^{\infty}$ or $\{a_n\}$，其中a_n称为第 n 项。

2. 数列极限

定义（ε-N 定义）：

$\lim\limits_{n\to\infty} a_n = L$ 表示，对于任意 ε>0，存在正整数 N，使得当 n≥N 时，$|a_n - L| < \varepsilon$。

3. 收敛与发散

收敛(convergence)：若 $\lim\limits_{n\to\infty} a_n = L$ 存在，则称 $\{a_n\}$ 收敛到 L。

发散(divergence)：若极限不存在（如振荡或趋于无穷大），则称数列发散。

例：$\lim\limits_{n\to\infty} [1 + \frac{(-1)^n}{n}] = 1$(收敛)

$\lim\limits_{n\to\infty} (-1)^n$ 不存在（发散）

4. 极限运算性质

若$\lim\limits_{n\to\infty} a_n = A$, $\lim\limits_{n\to\infty} b_n = B$, 则：

$\lim\limits_{n\to\infty} k = k$

$\lim\limits_{n\to\infty} ka_n = kA$

$\lim\limits_{n\to\infty} (a_n \pm b_n) = A \pm B$

$\lim\limits_{n\to\infty} (a_n \cdot b_n) = A \cdot B$

$\lim\limits_{n\to\infty} \frac{a_n}{b_n} = \frac{A}{B}$(若 $B \neq 0$)

5. 夹逼定理

若$a_n \le b_n \le c_n$(当 $n \ge N$)且 $\lim\limits_{n\to\infty} a_n = \lim\limits_{n\to\infty} c_n = L$，则$\lim\limits_{n\to\infty} b_n = L$。

6. 绝对值收敛定理

If $\lim\limits_{n\to\infty} |a_n| = 0$, then $\lim\limits_{n\to\infty} a_n = 0$.

7. 单调有界定理

单调递增且有上界（nondecreasing & bounded above）的数列必收敛。

单调递减且有下界（nonincreasing & bounded below）的数列必收敛。

【实战演练】

【Example1】

Determine whether the sequence converges or diverges. If it converges, find the limit:$a_n = \frac{(-1)^n\sqrt{n}+\ln n}{n+2}$

【Solution】

$$\lim_{n\to\infty}\frac{(-1)^n\sqrt{n}+\ln n}{n+2} = \lim_{n\to\infty}\frac{(-1)^n\sqrt{n}}{n+2} + \lim_{n\to\infty}\frac{\ln n}{n+2}$$

①for $\frac{(-1)^n\sqrt{n}}{n+2}$:

By Squeeze Theorem,

$0 \le \left|\frac{(-1)^n\sqrt{n}}{n+2}\right| \le \frac{\sqrt{n}}{n} = \frac{1}{\sqrt{n}} \to 0.$

Thus, $\lim_{n\to\infty}\frac{(-1)^n\sqrt{n}}{n+2} = 0.$

②for $\frac{\ln n}{n+2}$:

By L' Hospital's Rule, we have

$\lim_{n\to\infty}\frac{\ln n}{n+2} = \lim_{n\to\infty}\frac{1}{n} = 0.$

Conclusion: $\lim_{n\to\infty} a_n = 0 + 0 = 0.$

The sequence converges to 0.

【Example2】

Show that the sequence defined recursively by $a_1 = 4$, $a_{n+1} = \sqrt{6 + a_n}$ is convergent and find its limit.

【Solution】

Step1:Show the sequence is bounded and monotonic.

Boundedness:

By induction:

Base case:$a_1 = 4 \le 4$.

Assume $a_n \le 4$. Then,

$a_{n+1} = \sqrt{6 + a_n} \le \sqrt{6+4} = \sqrt{10} < 4$

Thus, the sequence is bounded above by 4.

Monotonicity:
Compute $a_{n+1} - a_n$:

$a_{n+1} - a_n = \sqrt{6 + a_n} - a_n$.

Let $f(x) = \sqrt{6 + x} - x$. For x> 0,

$f'(x) = \frac{1}{2\sqrt{6+x}} - 1 < 0$(since $\sqrt{6 + x} > 0$).

Since f(x) is decreasing and f(4)=$\sqrt{10} - 4 < 0$, we have $a_{n+1} < a_n$.

Thus, the sequence is monotonically decreasing.
Step2:Find the limit:

Let $\lim_{n\to\infty} a_n = L$. From the recursive formula:

$L = \sqrt{6 + L} \implies L^2 = 6 + L \Rightarrow L^2 - L - 6 = 0$.

Solve: $L = \frac{1\pm\sqrt{25}}{2}$ =3 or -2. Since $a_n > 0$, we discard -2.

Conclusion: The sequence converges to 3.

无穷级数

【基本思路】

1. 无穷级数: $\sum_{k=1}^{\infty} a_k$若 $\lim_{n\to\infty} S_n = S$ 存在，则级数收敛，和为 S, 否则发散。

2. 关键定理与检验法：

n 项检验法：若 $\lim_{n\to\infty} a_n \neq 0$，级数发散

几何级数：: $\sum_{k=1}^{\infty} ar^k$ 收敛当且仅当 ｜r｜<1，和为$\frac{a}{1-r}$

调和级数：: $\sum_{n=1}^{\infty} \frac{1}{n}$ 发散

坍塌级数（望远镜级数）：通过部分分式分解化简为可求和的级数。

3. 级数的性质：

收敛级数的线性组合仍收敛；发散级数乘以非零常数仍发散

【解题模板】

① 计算通项极限：计算 $\lim_{n\to\infty} a_n$

a. 若不为零，直接判定发散

b. 若为零，需进一步分析

②判断级数类型：

几何级数：形式为 $\sum_{k=1}^{\infty} ar^k$，公比 ｜r｜<1 时收敛

调和级数：形式为$\sum \frac{1}{n^p}$，当 p≤1 时发散。

坍塌级数：分解 $\frac{1}{(n+a)(n+b)}$, 依次消除求和。

③比较：

若通项可比较于已知收敛/发散级数，用比较法。

若级数绝对值通项趋于零，且正负交替，交错级数判别法

【实战演练】

【Example 1】

$$\sum_{n=1}^{\infty} \frac{n^3}{3n^3 + 2n^2}$$

【Solution】

Limit: $\lim_{n\to\infty} \frac{n^3}{3n^3+2n^2} = \frac{1}{3} \neq 0$，thus it diverges

【辅助理解】

通过 n 项检验法，判断通项极限不等于零，即发散

【Example 2】

$$\sum_{k=1}^{\infty}\frac{1}{(k+2)(k+3)}$$

【Solution】

$\frac{1}{(k+2)(k+3)} = \frac{1}{(k+2)} - \frac{1}{(k+3)}$,

Then

$S_n = (\frac{1}{3} - \frac{1}{4}) + \cdot\cdot\cdot + (\frac{1}{(n+2)} - \frac{1}{(n+3)}) = \frac{1}{3} - \frac{1}{(n+3)}$

$\lim_{n\to\infty} S_n = \frac{1}{3}$

Thus it converge, $\lim_{n\to\infty} S_n = \frac{1}{3}$

【辅助理解】

分解通项，求S_n依次消除，最后求极限$\lim_{n\to\infty} S_n$得到结论

正项级数：积分判别法

【基本思想】

1. 部分和有界判别法 Bounded Sum Test

当且仅当一个正项级数的部分和有上界，这个正项级数$\sum a_k$收敛。

（由 9.1TheoremD 延伸而来）

2.积分判别法 Integral Test

假设 f 在区间[1, ∞]上是连续(continuous)，正的(positive)，非递增(nonincreasing)函数，同时对于所有正整数 k 有$a_k = f(k)$, 那么当且仅当反常积分$\int_1^\infty f(x)dx$收敛时，无穷级数$\sum_{k=1}^\infty a_k$收敛。

（另一种表示方式：级数$\sum_{k=1}^\infty f(k)$和反常积分$\int_1^\infty f(x)dx$同时收敛或发散）

3. p 级数判别法 P-series Test

The p-series$\sum_{k=1}^\infty \frac{1}{k^p} = 1 + \frac{1}{2^p} + \frac{1}{3^p} + \frac{1}{4^p} + \dots \begin{cases} \text{converges, if } p > 1 \\ \text{diverges, if } p \le 1 \end{cases}$

4. Approximating the Sum of a Series 级数和的估算

关于求一个收敛级数的收敛值的问题，一般可以用积分判别法近似计算一个级数的和，例如调和级数的余项估算为$\int_n^\infty \frac{1}{x^3}dx = \frac{1}{2n^2}$，从而确定截断项数。

【实战演练】

【Example 1】

Decide the convergence or divergence of the series $\sum_{n=3}^\infty \frac{1}{n\ln n(\ln n)}$.

【Solution】

Use the Integral Test, $\int_3^\infty \frac{1}{x(\ln x)^2}dx = \ln 3$, the series converges.

拓展：

广义 p-级数 $\sum_{n=1}^{\infty}\frac{1}{n^{\alpha}(\ln n)^{\beta}}$, if $\begin{cases} \alpha > 1, \text{converges} \\ \alpha < 1, \text{diverges} \\ \alpha = 1, \begin{cases} \beta > 1, \text{converges} \\ \beta \le 1, \text{diverges} \end{cases} \end{cases}$

【Example 2】

Determine how large n must be so that using the nth partial sum to approximate the series$\sum_{k=n+1}^{\infty}\frac{1}{k^3}$gives an error of no more than10^{-4}(008 assignment 9-3)

【Solution】

$$R_n = \sum_{k=n+1}^{\infty}\frac{1}{k^3} \le \int_n^{\infty}\frac{1}{x^3}dx = [-\frac{1}{2x^2}]_n^{\infty} = \frac{1}{2n^2}$$

So that, $\frac{1}{2n^2} \le 10^{-4}$,

that is, n$\ge \sqrt{5000} \approx 70.71$

$$n = 71.$$

正项级数的判别法

【Theorem A】
一般形式的比较判别法（Ordinary Comparison Test）

Given two positive series $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$. If $\sum_{n=1}^{\infty} b_n$ converges and $a_n \le b_n$ then $\sum_{n=1}^{\infty} a_n$ **converges**. If $\sum_{n=1}^{\infty} b_n$ diverges and $a_n \ge b_n$ then $\sum_{n=1}^{\infty} a_n$ diverges.

【辅助理解】
此方法的要点是寻找一个收敛或发散的级数与原级数进行比较. 常用于比较的有如下两个级数：

等比级数：$\sum_{n=0}^{\infty} ar^n$，当|r|<1 时收敛，当|r| ≥ 1 时发散且$\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}$

p-级数 $\sum_{n=1}^{\infty} \frac{1}{n^p}$ 当 p > 1 时收敛，当 p ≤ 1 时发散

【Theorem B】
极限形式的比较判别法(Limit Comparison Test)

Let $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ **be two positive series, and** $\lim_{n\to\infty} \frac{a_n}{b_n} = L$

（1）if $0 < L < +\infty$ then $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ have the same convergence or divergence

（2）If $L = 0$ and $\sum_{n=1}^{\infty} b_n$ **converges then** $\sum_{n=1}^{\infty} a_n$ converges

（3）If $L = +\infty$ and $\sum_{n=1}^{\infty} b_n$ diverges，then $\sum_{n=1}^{\infty} a_n$ **diverges**.

【Theorem C】
比值判别法 (Ratio Test)

For a positive series $\sum_{n=1}^{\infty} a_n$，calculate $\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = \rho$

（1）If ρ < 1, the series converges.
（2）If $\rho > 1$ (**including** ρ = +∞), the series diverges.
（3）If $\rho = 1$, this method cannot determine the convergence or divergence of the series.

【实战演练】
【Example1】

Determine the convergence or divergence of the series $\sum_{n=1}^{\infty} \frac{1}{n^2+1}$

【Solution】

We know that the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is a convergent p- series (p = 2 > 1),and $\frac{1}{n^2+1} < \frac{1}{n^2}$, so $\sum_{n=1}^{\infty} \frac{1}{n^2+1}$ converges.

【Example2】

Determine the convergence or divergence of the series $\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^2+n}$

【Solution】

Take $b_n = \frac{1}{n^{\frac{3}{2}}}$ **and calculate** $\lim_{n\to\infty} \frac{\frac{\sqrt{n}}{n^2+n}}{\frac{1}{n^{\frac{3}{2}}}}$ = 1

The series $\sum_{n=1}^{\infty} \frac{1}{n^{\frac{3}{2}}}$ is a convergent p- series.

So $\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^2+n}$ converges.

【Example3】Determine the convergence or divergence of the series $\sum_{n=1}^{\infty} \frac{2^n}{n!}$

【Solution】

$$\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = \frac{2}{n+1} = 0 < 1$$

So $\sum_{n=1}^{\infty} \frac{2^n}{n!}$ converges

交错级数、绝对收敛和条件收敛

专题数学词汇： alternating series 交错级数
absolute convergence 绝对收敛
conditional convergence 条件收敛

【基本思想】

1. 交错级数定义 Definition：

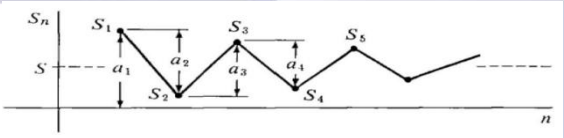
A series in which the terms are alternately positive and negative is an alternating series. For example,

$$a_1 - a_2 + a_3 - a_4 + a_5 - \dots + a_n - \dots$$

$$a_n = (-1)^{n+1}u_n \quad \text{or} \quad a_n = (-1)^n u_n$$
$$u_n = |a_n|$$

2. 交错级数的性质 Properties：

$$|S - S_n| \le a_{n+1}$$



3. 绝对收敛 Absolute Convergence：

Both the series $\sum_{n=1}^{\infty} a_n$ and the series $\sum_{n=1}^{\infty} |a_n|$ converge.

4. 条件收敛 Conditional convergence：

The series $\sum_{n=1}^{\infty} a_n$ converges, but the series $\sum_{n=1}^{\infty} |a_n|$ doesn't converge.

5. 绝对收敛定理 Absolute Convergence Test：

If $\sum_{n=1}^{\infty} |u_n|$ converges, then $\sum_{n=1}^{\infty} u_n$ converges.

【辅助理解】对于一般的级数 $\sum_{n=1}^{\infty} u_n$ 而言，若用正项级数审敛法判定 $\sum_{n=1}^{\infty} |u_n|$ 收敛，则 $\sum_{n=1}^{\infty} u_n$ 收敛。(常用于判断交错级数)

6. 绝对比率判别 Absolute Ratio Test

Let $\sum_{n=1}^{\infty} u_n$ be a series of nonzero terms and suppose that

$$\lim_{n\to\infty}\left|\frac{u_{n+1}}{u_n}\right| = \rho$$

(i) If ρ < 1, the series converges absolutely (hence converges).
(ii) If ρ > 1, the series diverges.
(iii) If ρ = 1, the test is inconclusive.

7.莱布尼茨定理：

If the alternating series $\sum_{n=1}^{\infty}(-1)^{n+1}a_n$ satisfies:

$$a_n \geq a_{n+1}$$

$$\lim_{n\to\infty} a_n = 0$$

the alternating series converges.

【辅助理解】

而且如果满足上述条件，其和$s_n \leq a_1$，其余和的绝对值$|r_n| \leq a_{n+1}$

8.交错 p 级数：

$$\sum_{n=1}^{\infty}(-1)^n\frac{1}{n^p} = \begin{cases} \text{absolute convergence} & p > 1 \\ \text{conditional convergence} & 0 < p \leq 1 \\ \text{divergence} & p < 0 \end{cases}$$

【解题模板】

交错级数：

1. 绝对收敛定理解决，判断$\sum_{n=1}^{\infty}|u_n|$
2. 绝对比率判别,判断 $\lim_{n\to\infty}\left|\frac{u_{n+1}}{u_n}\right| = \rho$
3. 或者运用莱布尼茨定理

判断绝对收敛还是条件收敛

1. 直接法：判断 $\sum_{n=1}^{\infty}|a_n|$
2. 绝对比率判别 $\lim_{n\to\infty}\left|\frac{u_{n+1}}{u_n}\right| = \rho$
3. 交错 p 级数

【Example 1】

Determine the convergence or divergence of the series:

$$\sum_{n=1}^{\infty}(-1)^{n+1}\frac{n^2}{e^n}$$

【Solution】

Suppose $u_n = \frac{n^2}{e^n}$

$$\lim_{n\to\infty}\left|\frac{u_{n+1}}{u_n}\right| = \lim_{n\to\infty}\frac{(n+1)^2}{n^2}\cdot\frac{1}{e} = \lim_{n\to\infty}\left(1+\frac{2}{n}+\frac{1}{n^2}\right)\cdot\frac{1}{e} = \frac{1}{e} < 1$$

the series converges absolutely.

【Example 2】

Determine the convergence or divergence of the series:

$$\sum_{n=1}^{\infty}(-1)^{n+1}\sqrt{n}\,(1-\cos\frac{1}{n})$$

【Solution】

Let $u_n = (-1)^{n+1}\sqrt{n}\,(1-\cos\frac{1}{n})$ and $b_n = \frac{1}{n^{\frac{3}{2}}}$

As $(n\to\infty)$ $1-\cos\frac{1}{n} \sim \frac{1}{2n^2}$, we have

$$\lim_{n\to\infty}\frac{|u_n|}{b_n} = \lim_{n\to\infty}\frac{\sqrt{n}\,(1-\cos\frac{1}{n})}{\frac{1}{n^{\frac{3}{2}}}} = \frac{1}{2} > 0$$

Besides, the series $\sum_{n=1}^{\infty} b_n$ converges.

So, the series $\sum_{n=1}^{\infty}|u_n|$ converges.

The series $\sum_{n=1}^{\infty} u_n$ converges absolutely.

【Example 3】

Determine the convergence or divergence of the series:

$$\sum_{n=1}^{\infty}\frac{\cos n\pi}{n}$$

【Solution】

Let $u_n = \frac{\cos n\pi}{n}$

And we can find that $u_n = \frac{\cos n\pi}{n} = (-1)^{n+1}\cdot\frac{1}{n}$

Then, the series $\sum_{n=1}^{\infty}|u_n|$ diverges.

Since $\lim_{n\to\infty}|u_n| = 0$ and $|u_n| > |u_{n+1}|$

Thus the series is conditional convergent.

幂级数

【基本思想】
1.幂级数的定义（Power Series Definition）
幂级数是以 x-a 为中心的无穷级数，形式为：

$$\sum_{n=0}^{\infty} a_n (x-a)^n$$

当 a=0 时，简化为$\sum_{n=0}^{\infty} a_n x^n$。例如：$\sum_{n=0}^{\infty} a_n \frac{x^n}{n!}$是幂级数，其和为$e^x$。

2.收敛集与收敛半径(Convergence & Radius of Convergence)
收敛集是使幂级数收敛的所有 x 值的集合，可能为以下三种类型：
①单点 x=a（收敛半径 R=0）。
②区间（a−R, a+R），可能包含端点（收敛半径 R>0）。
③全实数轴（−∞,∞）（收敛半径 R=∞）。

收敛半径 R 通过比值测试计算：

$$\rho = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| \Rightarrow R=\frac{1}{\rho}$$

3.收敛集的性质（Properties of Convergence Set）
幂级数在收敛区间内部（即|x-a|<R）绝对收敛。
端点 x=a±R 需要单独测试（如交替级数测试、比较测试）。
收敛区间对称于中心点 a。

【实战演练】
【Example 1】
Find the convergence set for the power series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{\sqrt{n}}(x-1)^n$

【辅助理解】
使用比值测试确定收敛半径 R 与收敛区间
端点逐个带入检查，只要是收敛就需要是闭区间
【Solution】
Step 1:Apply the Ratio Test

$$\rho = \lim_{n\to\infty} \left|\frac{(x-1)^{n+1}}{\sqrt{n+1}} \cdot \frac{\sqrt{n}}{(x-1)^n}\right| = |x-1| \lim \sqrt{\frac{n}{n+1}} = |x-1|$$

The series converges absolutely if ρ<1, i.e., |x-1|<1, so the radius of the convergence R=1.

Step2: Check Endpoints

①x=2：

The series becomes $\sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{\sqrt{n}}$, which converges conditionally by the Alternating Series Test.

②x=0:

The series becomes $\sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{\sqrt{n}}(-1)^n=\sum_{n=1}^{\infty}\frac{-1}{\sqrt{n}}$, which diverges conditionally by the Alternating Series Test.

Conclusion:

The convergence set is (0,2].

【Example 2】

Determine the radius and interval of convergence for $\sum_{n=1}^{\infty} n!\,(x)^n$

【辅助理解】

比值极限为无穷大，说明除 x=0 外均发散。收敛集仅为单点{0}。

【Solution】

Apply the Ratio Test

$$\rho=\lim_{n\to\infty}\left|\frac{(n+1)!x^{n+1}}{n!x^n}\right|=\lim_{n\to\infty}(n+1)|x|=\infty \text{ (if } x\neq 0)$$

Conclusion:

The series converges only at x=0.

幂级数运算

【基本思想】

1. 若 S(x)既可微又可积，那么其导数和积分可以通过逐项微分积分来计算。

Theorem A 逐项微分和积分(Term-by-Term Differentiation and Integration)：

假设 S(x)是一个幂级数在区间 I 上的和，即

$$S(x)=\sum_{n=0}^{\infty}a_nx^n=a_0+a_1x+a_2x^2+a_3x^3+\cdots$$

那么，当 x 是 I 的内点时有

$$\text{(i)}\quad S'(x)=\sum_{n=0}^{\infty}D_x\left(a_nx^n\right)=\sum_{n=1}^{\infty}n\,a_nx^{n-1}=a_1+2a_2x+3a_3x^2+\cdots$$

$$\text{(ii)}\ \int_0^x S(t)dt=\sum_{n=0}^{\infty}\int_0^x a_n\,t^ndt=\sum_{n=0}^{\infty}\frac{a_n}{n+1}x^{n+1}$$

$$=a_0x+\frac{1}{2}a_1x^2+\frac{1}{3}a_2x^3+\frac{1}{4}a_3x^4+\cdots$$

我们可以将 Theorem A 应用于已知和的幂级数，以获得其他序列的和公式；或者找到某些函数的幂级数表示。

2. 判断幂级数在收敛集的端点的敛散性

导数和积分序列的收敛半径 R 与原序列相同，但在区间端点处，导数和积分序列的收敛性和发散性可能会改变。

假设

$$f(x)=\sum_{n=0}^{\infty}a_nx^n,\ |x|<R$$

如果函数 f 在收敛集的端点 (R 或-R) 连续且级数在端点处也收敛，则公式在端点处保持收敛的性质。

3.Theorem B

令 f(x)= $\sum a_nx^n$和 g(x)=$\sum b_nx^n$，并且这两个级数至少对于$|x|<r$ 收敛。如果对这两个级数像多项式那样进行加减运算，得到的级数 f(x)+g(x), f(x)-g(x)及 f(x)*g(x)，对于$|x|<r$ 也分别收敛。如果$b_0\neq 0$，那么对于除法也能得出相应的结论，但我们只能保证它仅对充分小的$|x|$有效。

【解题模板】求出和函数的幂级数

①步骤一：

对于给定的函数，尝试找到一个幂级数，该级数在某区间内收敛到该函数。（通常涉及到泰勒级数）

②步骤二：

利用得到的幂级数来解决具体问题（计算极限、求解积分、研究函数的性质等）

③步骤三（**特别注意**）：
对于闭区间，还需要检查端点处的收敛性，因为即使在区间内部收敛，在端点处也可能发散。
【实战演练】
【Example 1】

Find the power series for $e^{\tan^{-1}x}$

【辅助理解】
涉及重要的麦克劳林级数e^x

【Solution】

$$e^u = 1 + u + \frac{u^2}{2!} + \frac{u^3}{3!} + \frac{u^4}{4!} + \cdots$$

$$e^{\tan^{-1}x} = 1 + \tan^{-1}x + \frac{(\tan^{-1}x)^2}{2!} + \frac{(\tan^{-1}x)^3}{3!} + \frac{(\tan^{-1}x)^4}{4!} + \cdots$$

$$e^{\tan^{-1}x} = 1 + \left(x - \frac{x^3}{3} + \cdots\right) + \frac{\left(x - \frac{x^3}{3} + \cdots\right)^2}{2!} + \frac{\left(x - \frac{x^3}{3} + \cdots\right)^3}{3!} + \frac{\left(x - \frac{x^3}{3} + \cdots\right)^4}{4!} + \cdots$$

【Example 2】

Find the power series in x for function $f(x) = \frac{\tan^{-1}x}{e^x}$

【Solution】

$$= e^{-x}\tan^{-1}x$$

$$= \left(1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \cdots\right)\left(x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots\right)$$

$$= x - x^2 + \frac{x^3}{6} + \frac{x^4}{6} + \frac{3x^5}{40} + \cdots$$

求麦克劳林级数

【基本思路】

1. 泰勒级数：a 的幂级数展开：f(x)= $\sum_{n=0}^{\infty}\frac{f^{(n)}(a)}{n!}(x-a)^n$

2. 麦克劳林级数：泰勒级数在 a=0 时的特例，形式为：f(x)=$\sum_{n=0}^{\infty}\frac{f^{(n)}(0)}{n!}x^n$

3. 第 n 项系数：$C_n=\frac{f^{(n)}(a)}{n!}$

4. 唯一性定理：
   若函数 f 能表示为幂级数，则该级数必为泰勒级数。

5, 余项随着 n 增大趋近于零是泰勒级数收敛到原函数的充要条件

   其中余项$R_n(x)=\sum_{n=0}^{\infty}\frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}$

6. 重要函数的麦克劳林级数展开式

(1) $e^x=1+x+\frac{x^2}{2!}+\cdots+\frac{x^n}{n!}+o(x^n)$

(2) $\sin x=x-\frac{x^3}{3!}+\frac{x^5}{5!}-\frac{x^7}{7!}+\cdots+(-1)^n\frac{x^{2n+1}}{(2n+1)!}+o\left(x^{2n+2}\right)$

(3) $\cos x=1-\frac{x^2}{2!}+\frac{x^4}{4!}-\frac{x^6}{6!}+\cdots+(-1)^n\frac{x^{2n}}{(2n)!}+o\left(x^{2n+1}\right)$

(4) $\ln(1+x)=x-\frac{x^2}{2}+\frac{x^3}{3}-\frac{x^4}{4}+\cdots+(-1)^n\frac{x^{n+1}}{n+1}+o\left(x^{n+1}\right)$

(5) $\frac{1}{1-x}=1+x+x^2+\cdots+x^n+o(x^n)$ [2]

(6) $(1+x)^\alpha=1+\alpha x+\frac{\alpha(\alpha-1)}{2!}x^2+\cdots+\frac{\alpha(\alpha-1)\cdots(\alpha-n+1)}{n!}x^n+o(x^n)$

【解题模板】

①：求导数

计算函数 f（x）在 a 处的各阶导数$f^{(n)}(a)$

②：写通项公式

代入泰勒级数通项：$C_n=\frac{f^{(n)}(a)}{n!}$

③：写出级数形式

将系数代入泰勒级数：$f(x)=\sum_{n=0}^{\infty}\frac{f^{(n)}(a)}{n!}(x-a)^n$

④：验证余项收敛

证明$\lim_{n\to\infty}R_n(x)=0$常用方法：

利用$|R_n(x)|\leqslant\frac{M}{(n+1)!}|x-a|^{n+1}$

若 $\frac{|x-a|^{n+1}}{(n+1)!} \to 0$，则余项趋近于 0

【实战演练】

Example1

Find the maclaurin series for sinx and prove that it represent sinx for all x

Solution

f(x) = sinx , f(0) =0

f'(x) =cosx , f'(0) =1

f''(x) = -sinx , f''(0) = 0

f'''(x) = -cosx , f'''(0) = -1

Since $|f^{(n+1)}(c)|$ = |cosx| or |sinx|

$$|R_n(x)| = \frac{f^{(n+1)}(c)}{(n+1)!}|x|^{n+1} \leqslant \frac{|x|^{n+1}}{(n+1)!}$$

and $\lim_{n\to\infty} \frac{|x|^n}{(n)!} = 0$ for all x

thus $\lim_{n\to\infty} R_n(x) = 0$ for all x

$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots \quad -\infty < x < \infty$

【辅助理解】

依次求导得到$f^{(n)}(x)$，代入通项$\frac{f^{(n)}(0)}{n!}x^n$得到结果。通过放缩得到余项趋近于零对任意 x 都成立，级数收敛

【Example 2】

Find the Maclaurin series for f(x)= ln(1+x),also find the associated radius of covergence

【Solution】

f(x) = ln(1+x) , f(0) =0

f'(x) =$(1+x)^{-1}$, f'(0) =1

f''(x) = $-(1+x)^{-2}$, f''(0) = -1

f'''(x) = $2(1+x)^{-3}$, f'''(0) = 2

$$\ln(1+x) = f(0) + f'(0)x + f''(0)\frac{x^2}{2!} + f'''(0)\frac{x^3}{3!} + \cdots$$

$$= 0 + x - \frac{x^2}{2} + \frac{2x^3}{6} - \frac{6x^4}{24} + \cdots$$

$$= x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$$

$$= \sum_{n=1}^{\infty} \frac{-1^{(n-1)}}{n} x^n$$

$\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \left|\frac{x^{n+1}}{n+1} \cdot \frac{n}{x^n}\right| = \lim_{n\to\infty} \frac{|x|}{1+\frac{1}{n}} = |x| < 1$ for convergence,

So R = 1